

Touchscreen and Multi-Function Make Today's Oscilloscopes Powerful



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Displaying variations in current and voltages with respect to time and even frequency, multi-function oscilloscopes have been around for a long time. These oscilloscopes are vital for, and most commonly used by, professional engineers, do-it-yourselfers (DIYers) and hobbyists who are into manufacturing, designing or repairing. Scientists and researchers working on next-generation systems and components use oscilloscopes with ultra-high bandwidths.

Oscilloscopes have always been the tool of choice for engineers debugging electronic devices. Most critical parameters for verifying digital designs are time related, informs Trevor J. Smith, business develop-

ment manager, Pico Technology. The next generation of mixed-signal designs employ a broad range of digital and analogue signals including all sorts of sensors, smart power controls, audio, video and more, which make required oscilloscope specifications more challenging. He says, "In addition to timing parameters, users need to make accurate measurements of such characteristics as total harmonic distortion (THD), crosstalk, and signal-to-noise and distortion ratio (SINAD)."

Over the years, multi-function oscilloscopes have undergone many transformations, be it with respect to features or technology. In addition to latest happenings and trends surrounding these oscilloscopes, here we also highlight some modern instruments in the market that meet today's demanding measurement challenges.

Top feature highlights in the last one year

- Capacitive touchscreens looking to replace knob based operation
- Integration of multiple test and measurement tools in one instrument
- Ultra-high bandwidths ranging up to 100GHz for greater insights for validation and debugging
- Oscilloscopes for analysis of specific designs using low-power double-data-rate 4 memory (LPDDR4), pulse amplitude modulation with four amplitude levels (PAM-4) signals, USB 3.1, MIPI M-PHY serial interface technology and CAN flexible data (CAN FD) interface

Point of view: Emergence of alternate operating systems for software applications bundled along with multi-function oscilloscopes

When at home, many of us swap between tablets, phones and laptops without thinking about different operating systems these are using. So far, Windows-powered PCs have remained dominant in the average lab. However, this is slowly changing. A reluctance to upgrade to Windows 8 and greater familiarity of alternatives is leading to increasing demands in being able to run design tools and test equipment on multiple platforms such as Linux, Android and Mac OS X.

As manufacturers respond to these demands, dominance of Windows is being challenged. For some applications, tablets are being used alongside Windows PCs. In other cases, products such as iMacs and MacBooks, which were previously only seen in sales and marketing departments, are finding their way into the lab as replacements for Windows PCs.

—Trevor J. Smith, business development manager, Pico Technology

Latest multi-function oscilloscopes aim to boost testing process

Latest oscilloscopes come with features that help engineers solve problems quicker and, in turn, improve overall productivity. We throw light on some notable features below.

Touchscreen, knob-free capabilities.

Capacitive touchscreens are not just limited to industrial test and measurement (T&M) instruments today. T&M vendors are slowly bringing touchscreens to mainstream oscilloscopes as well. This feature would assist engineers overcome usability challenges. In fact, some vendors are manufacturing full-touchscreen scopes.

For the first time Keysight Technologies has introduced touchscreen capability in its 3000T X-series of digital and mixed-signal oscilloscopes. This series of oscilloscopes was particularly designed for operation with capacitive touchscreens that permit engineers to pick targets naturally and swiftly.